MILITARY SPECIFICATION SHEET

CONTACT, ELECTRICAL CONNECTOR
CONCENTRIC TWINAX, PIN, SIZE 8
(FOR DOD-C-83527 CONNECTOR)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The complete requirements for acquiring the contact described herein shall consist of this specification and the latest issue of MIL-C-39029.

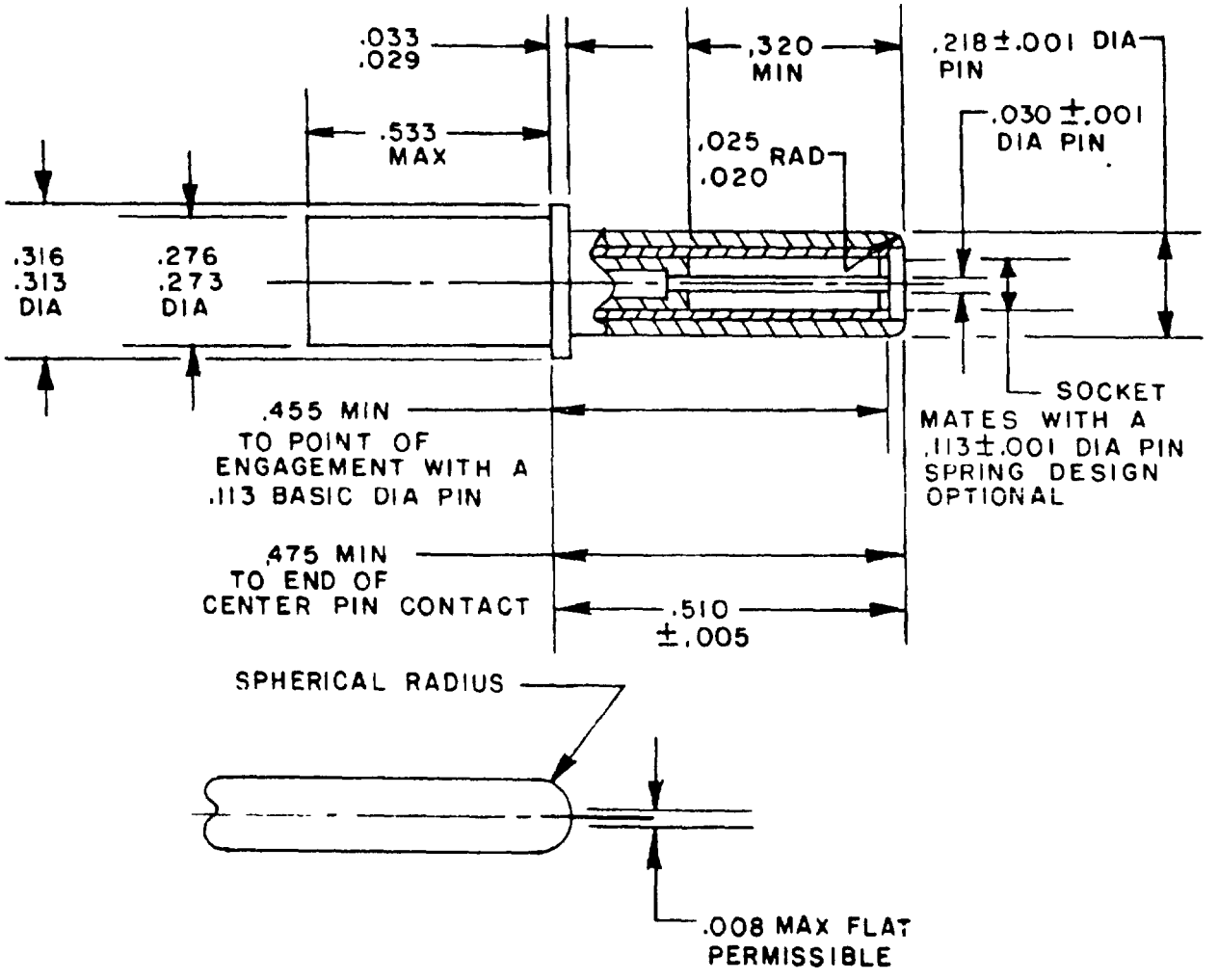


FIGURE 1 Concentric twinax contact

FSC 5935

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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.001	0.03	029	0.74	.273	6.93	.455	11.56
005	0 13	030	0.76	.276	7.01	475	12 07
008	0 20	033	0.84	313	7 95	.510	12.95
.020	0.51	.113	2.87	.316	8.03	.533	13 54
.025	0 64	218	5.54	.320	8.13		

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.

FIGURE 1. Concentric twinax contact - Continued.

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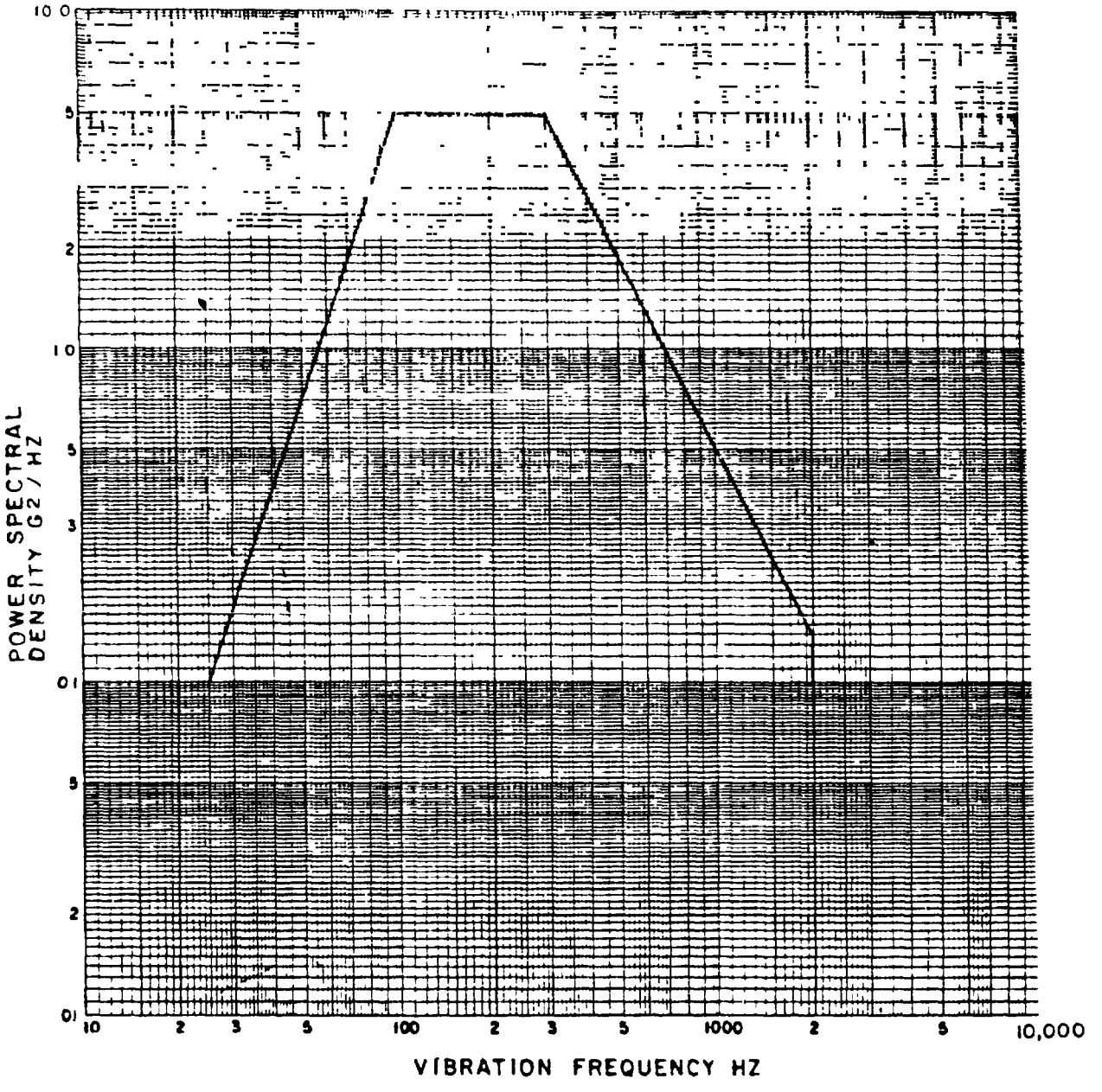


FIGURE 2. Vibration envelope.

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ENGINEERING DATA

Frequency range 0 to 20 MHz.

Voltage rating 500 volts rms maximum, working voltage at sea level, 125 volts rms maximum at 70,000 feet.

Temperature rating -65°C to +150°C.

REQUIREMENTS.

Design and construction

Dimensions and configuration See figure 1 and table I

Tools See tables II and III

Assembly procedure Assembly instructions to be developed. Manufacturer's recommended assembly instructions shall be shipped with unit package.

Mating contact See MIL-C-39029/95.

Low signal level contact resistance (center contact only) See table IV

Contact resistance. See table V.

Contact engagement and separation force. The engagement depth shall be as encountered in normal service. The test pins shall be in accordance with MS3197, except the diameters shall be as specified in table VI

Crimp tensile strength (center, intermediate, and outer contact crimp joint) See table VII

Vibration Method 2005 of MIL-STD-1344, test condition V The following details shall apply

a Use the vibration envelope shown on figure 2 (derived from zone 2 outlined in Aerospace information report AIR 1557).

b. Vibration To be conducted at standard test conditions

c. Duration 8 hours in the longitudinal direction and 8 hours in a perpendicular direction (16 hours).

High impact shock All contacts shall be wired in a series circuit with 100 milliamperes maximum current flow through the series circuit during shock Connectors shall be monitored for any discontinuities A detector capable of detecting all discontinuities in excess of 1 microsecond shall be used. Wired and mated connectors shall be subjected to the test specified in MIL-S-901, grade A with the following modifications and additions: Mounting fixture shall be in accordance with MIL-S-901, lightweight. The cable or wire bundle shall be supported on a stationary frame to provide a free flexing cable length between the frame and fixture of not less than 36 inches (914 4 mm)

a Coupling Coupled by normal mounting means.

b Test conditions A Plug shall be terminated with at least 80 percent of wired contacts The wire bundle shall be provided with straight, open frame, strain relief accessory hardware.

Insulation resistance Applicable.

Corrosion (salt spray) Applicable

Humidity Method 1002, type II, per MIL-STD-1344

Dielectric withstanding voltage See table VIII

Durability Applicable.

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Thermal shock High temperature shall be +85°C except for connectors using +200°C cables, high temperature shall then be +150°C.

Permeability Applicable.

Temperature life Applicable.

Plating porosity Applicable.

Part number M39029/96-546.

QPL evaluating activity Defense Electronics Supply Center (DESC-E), Dayton, OH 45444.

Contacts shall comply with reliability assurance provisions of MIL-STD-790 as specified in MIL-C-38999.

TABLE I. Design characteristics

BIN code	Color bands			Contact cavity size	Cable accommodated	Type	Class
	1st	2nd	3rd				
546	Green	Yellow	Blue	8	M17/176-00002	E	B

TABLE II. Tools (installing and removal).

BIN code	Installing and removal tool
546	M81969/14-06

TABLE III. Tools (crimping).

BIN code	Center contact tooling		Intermediate contact tooling		Outer crimp sleeve tooling	
	Basic crimping tool	Contact positioner	Basic crimping tool	Contact positioner	Basic crimping tool	Die part no.
546	M22520/XX 1/	M22520/XX 1/	M22520/XX 1/	M22520/XX 1/	M22520/XX M22520/XX 1/	M22520/XX Die M22520/XX 1/

1/ To be determined.

TABLE IV. Low signal level contact resistance (center contact only).

BIN code	Cable accommodated	Maximum contact resistance (milliohms)	
		Initial	After conditioning
546	M17/176-00002	55	66

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TABLE V Contact resistance.

BIN code	Contact	Cable accomodated	Test current (amperes)	Maximum voltage drop (millivolts)		
				25°C, +3°C, -0°C	+200°C, +3°C, -0°C	
				Initial	After conditioning	After conditioning
546	Center	M17/176-00002	1.0	55	66	94
	Intermediate		1.0	55	66	94
	Outer		12.0	75	90	126

TABLE VI Contact engagement and separation force.

Test pin diameter (inch)	Minimum separation force (ounces)		Maximum engagement force (ounces)		Maximum average engagement force
	Initial	After conditioning	Initial	After conditioning	
.1140 +.0000 -.0002	NA	NA	30	36	NA
.1120 +.0002 -.0000	3.0	2.5	NA	NA	NA

TABLE VII. Crimp tensile strength.

BIN code	Cable accommodated	Axial load (pounds, minimum)
546	M17/176-00002	25

TABLE VIII. Dielectric withstanding voltage

Contacts	Altitude	Test voltage ac rms
Center to intermediate	Sea level	1000
Intermediate to outer	Sea level	500

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Custodians
  Army - CR
  Navy - AS
  Air Force - 85

Review activities.
  Navy - EC, SH, MC
  Air Force - 11, 15, 17, 99
  DLA - ES

User activities
  Army - AT
  Navy - OS

Agent·
  DLA - ES

Preparing activity
  Air Force - 85

(Project 5935-3415-4)

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